

BUILDING A VIRTUAL SOC ENVIRONMENT

Cybersecurity Project Report

Tools: Splunk • Wazuh • Windows 11 • Ubuntu • VirtualBox

Prepared by: Vishal Bhambhana

1. Project Overview

The objective of this project is to build a small Virtual Security Operations Center (SOC) environment for collecting security logs, monitoring endpoint activity, detecting suspicious events, investigating alerts, and demonstrating basic incident response.

The lab uses Splunk for log collection and analysis and Wazuh for endpoint monitoring, file integrity monitoring, and security event detection. A Windows 11 endpoint is used to generate controlled security events for testing.

2. Lab Environment

The lab consists of the following components:

Component	Purpose	Platform
Splunk Server	Log collection, search, analysis and dashboarding	Ubuntu VM
Wazuh Manager	Endpoint monitoring and security event detection	Wazuh OVA
Windows Endpoint	Source of Windows security events	Windows 11
Virtualization	Runs the lab virtual machines	VirtualBox

3. Splunk Setup and Log Collection

Splunk was configured on the Ubuntu virtual machine. The Windows 11 endpoint was connected using the Splunk Universal Forwarder so that Windows event logs could be forwarded to Splunk.

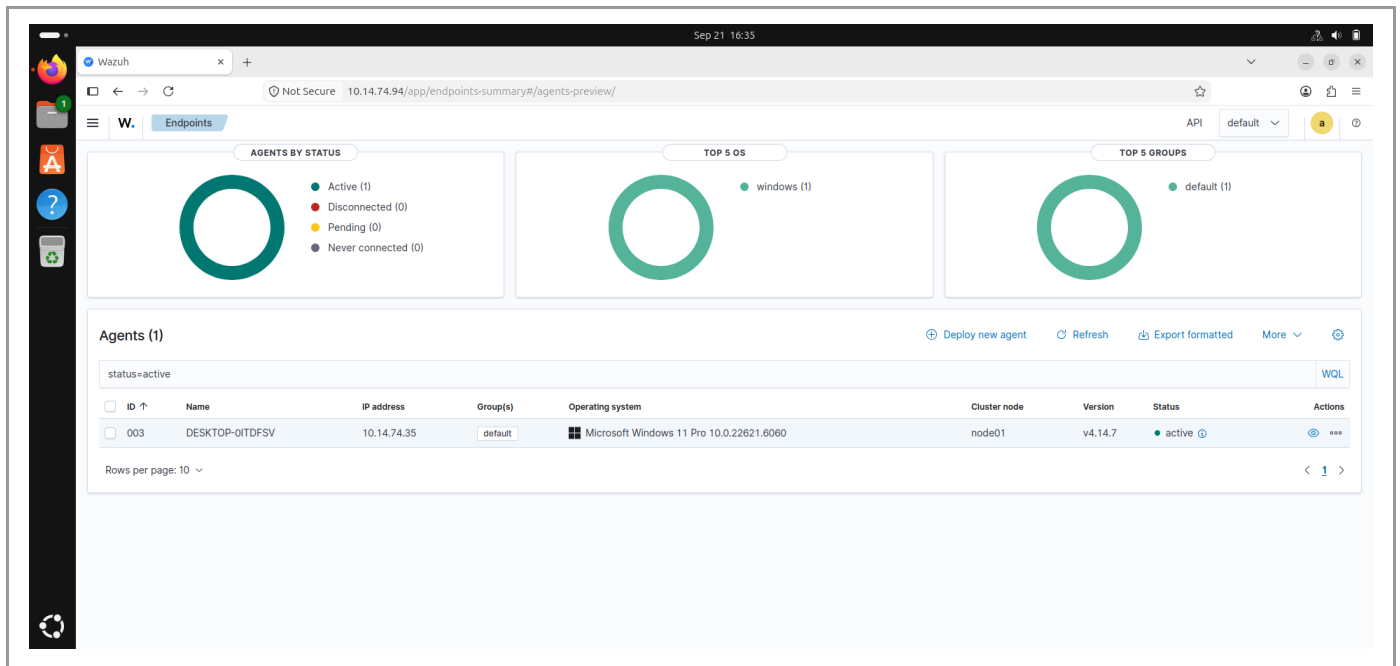
The forwarding connection was verified by searching recent events in Splunk. Windows Security events were available for investigation and dashboard creation.

The screenshot displays the Splunk Enterprise web interface. The top navigation bar shows the user is logged in as 'Administrator'. The left sidebar contains various navigation options: Search & Reporting, Search, Analytics, Datasets, Reports, Alerts, Dashboards, and Modules. The main content area is titled 'New Search' and shows a search query 'index=' with a search button. Below the search bar, it indicates '4,167 events' for the time range '9/23/26 2:00:00.000 AM to 9/24/26 2:48:40.000 AM'. The search results are displayed in a table format, showing columns for Time and Event. The first event is from 9/24/26 at 2:45:03.939 AM, with LogName=Security, EventCode=5379, EventType=0, and ComputerName=DESKTOP-0ITDFSV. The second event is from 9/24/26 at 2:44:51.880 AM, with LogName=Security, EventCode=5379, EventType=0, and ComputerName=DESKTOP-0ITDFSV. The third event is from 9/24/26 at 2:44:51.880 AM, with LogName=Security, EventCode=5379, EventType=0, and ComputerName=DESKTOP-0ITDFSV. The interface also includes a 'Selected Fields' list on the left and a 'View: List' option at the bottom right.

4. Wazuh Setup and Endpoint Monitoring

Wazuh Manager was deployed using the Wazuh virtual appliance. The Windows 11 endpoint was enrolled as a Wazuh agent. The agent was verified as active and connected to the Wazuh manager.

Wazuh was also configured for real-time File Integrity Monitoring (FIM) on the Windows Public folder.



5. Incident 1 — Failed Windows Login

Detection

A controlled failed-login scenario was generated on the Windows 11 endpoint by entering an incorrect password several times. Windows generated Security Event ID 4625.

Splunk detected EventCode 4625. Wazuh detected the failed authentication activity using Rule ID 60122, described as a logon failure caused by an unknown user or bad password.

Evidence collected:

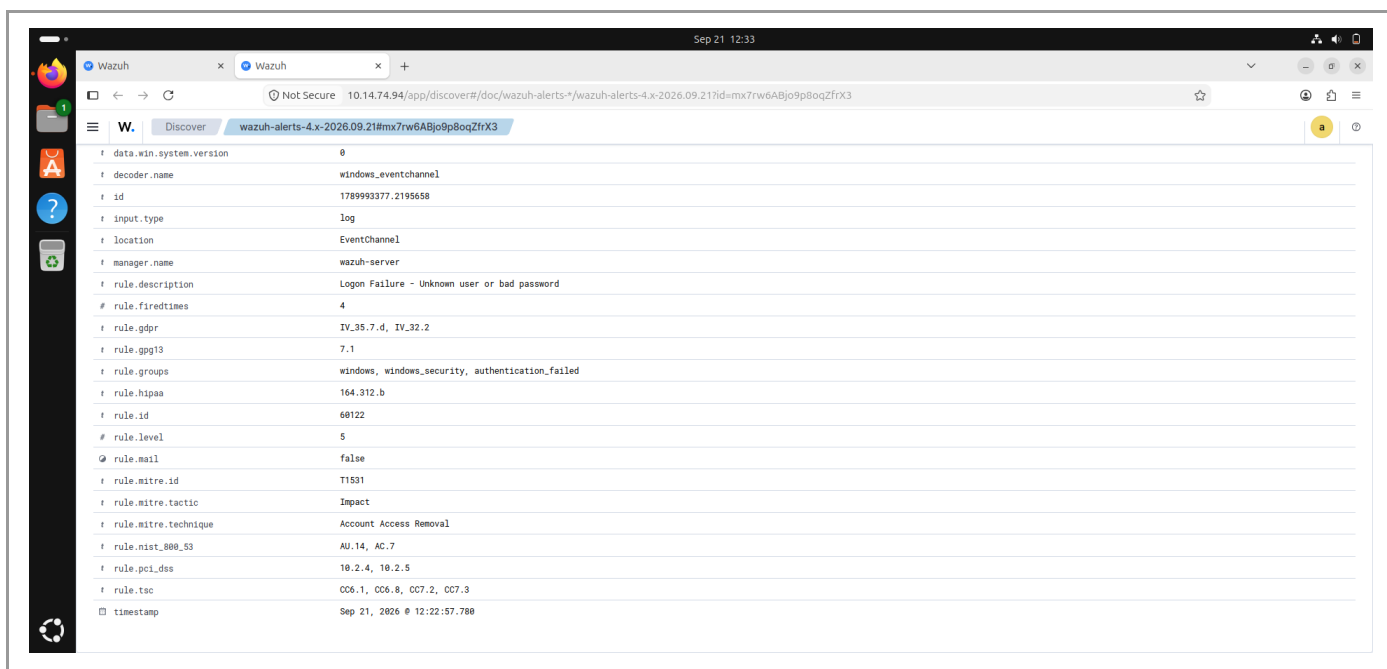
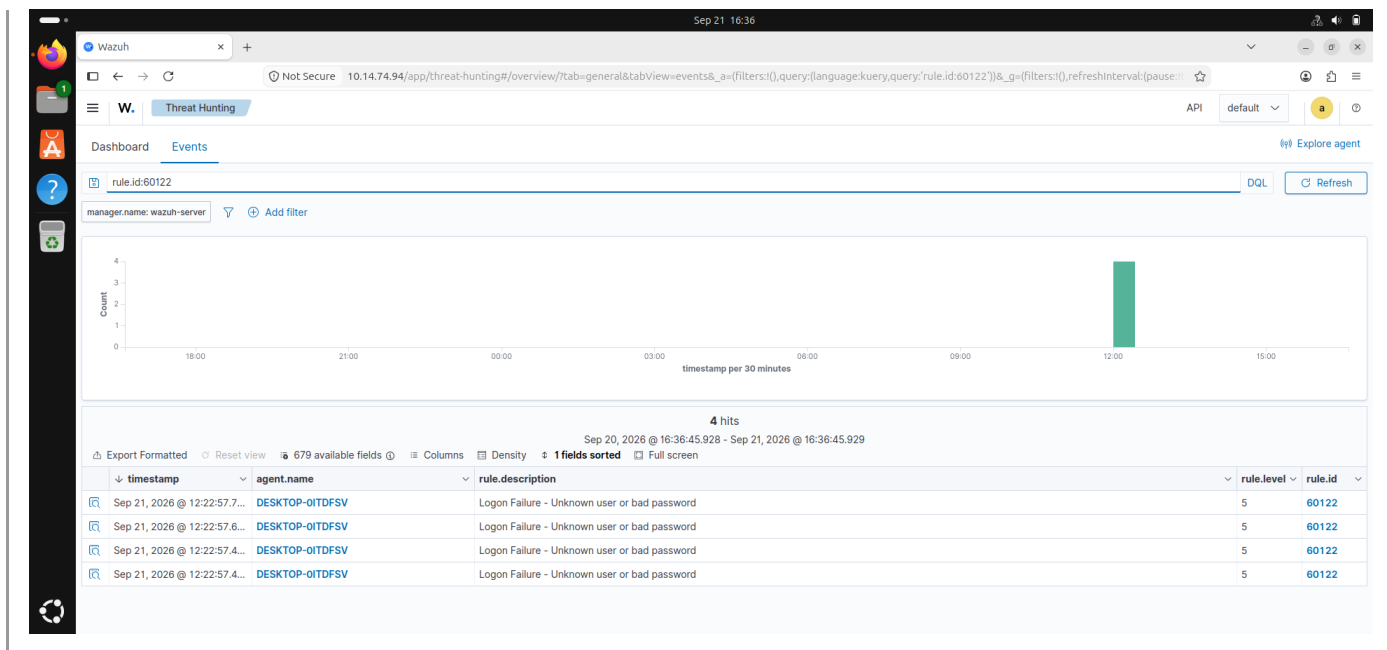
- Windows Event ID 4625
- Splunk failed-login search result
- Wazuh Rule ID 60122
- Wazuh failed-login event details

Splunk Enterprise interface showing a search for EventCode=4625. The search results are displayed in a table format, showing the Time, Event, and LogName=Security. The search criteria are: Index=* EventCode=4625 earliest=-24h. The results show three events, all with EventCode=4625 and Event Type=0. The first event is from 09/21/2026 05:48:25.557 PM, the second from 09/21/2026 05:48:20.086 PM, and the third from 09/21/2026 05:48:16.032 PM. The source is WinEventLog:Security and the sourcetype is WinEventLog:Security.

Time	Event
09/21/2026 05:48:25.557 PM	LogName=Security EventCode=4625 Event Type=0 ComputerName=DESKTOP-GITDFSV Show all 61 lines
09/21/2026 05:48:20.086 PM	LogName=Security EventCode=4625 Event Type=0 ComputerName=DESKTOP-GITDFSV Show all 61 lines
09/21/2026 05:48:16.032 PM	LogName=Security EventCode=4625 Event Type=0 ComputerName=DESKTOP-GITDFSV Show all 61 lines

Splunk Enterprise interface showing the expanded search results for EventCode=4625. The search criteria are: Index=* EventCode=4625 earliest=-24h. The results show three events, all with EventCode=4625 and Event Type=0. The first event is from 09/21/2026 05:48:25.557 PM, the second from 09/21/2026 05:48:20.086 PM, and the third from 09/21/2026 05:48:16.032 PM. The source is WinEventLog:Security and the sourcetype is WinEventLog:Security. The expanded view shows the following fields:

Type	Field	Value
Selected	host *	DESKTOP-GITDFSV
	source *	WinEventLog:Security
	sourcetype *	WinEventLog:Security
Event	Account_Domain *	WORKGROUP
	Account_Name *	DESKTOP-GITDFSV
	Authentication_Package *	Negotiate
	Caller_Process_ID *	0x80
	Caller_Process_Name *	C:\Windows\System32\svchost.exe
	ComputerName *	DESKTOP-GITDFSV
	EventCode *	4625
	Event Type *	0
	Failure_Reason *	An error occurred during login.
	Key_Length *	0
	Keywords *	Auth Failure
	LogName *	Security
	Logon_ID *	0x3e7
	Logon_Process *	User32
	Logon_Type *	2
	Message *	An account failed to log on. Subject: Security ID: S-1-5-18 Account Name: DESKTOP-GITDFSV Account Domain: WORKGROUP Logon ID: 0x3e7 Logon Type: 2 Account For Which Logon Failed: Security ID: S-1-0-0 Account Name: - Account Domain: - Failure Information: Failure Reason: An error occurred during login. Status: 0xc000006d Sub Status: 0xc0000080 Process Information: Caller Process ID: 0x80 Caller Process Name: C:\Windows\System32\svchost.exe Network Information: Workstation Name: - Source Network Address: 127.0.0.1 Source Port: 0 Detailed Authentication Information: Logon Process: User32 Authentication Package: Negotiate Translated Services: - Package Name (NTLM only): - Key Length: 0 This event is generated when a login request fails. It is generated on the computer where access was attempted. The Subject fields indicate the account on the local system which requested the login. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe. The Logon Type field indicates the kind of login that was requested. The most common types are 2 (interactive) and 3 (network). The Process Information fields indicate which account and process on the system requested the login. The Network Information fields indicate where a remote login request originated. Workstation name is not always available and may be blank in some cases. The authentication information fields provide detailed information about this specific login request. Translated services indicate which intermediate services have participated in this login request. - Package name indicates which sub-protocol was used among the NTLM protocols. - Key length indicates the length of the generated session key. This will be 0 if no session key was requested.

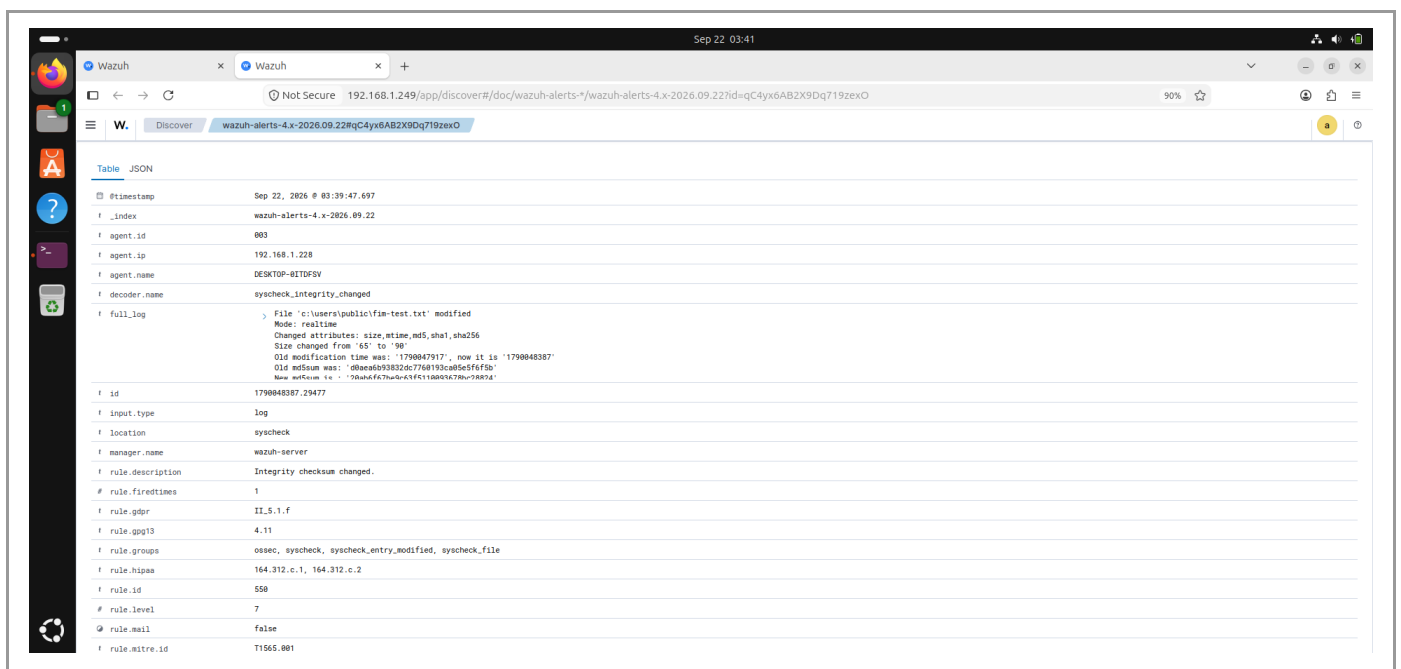
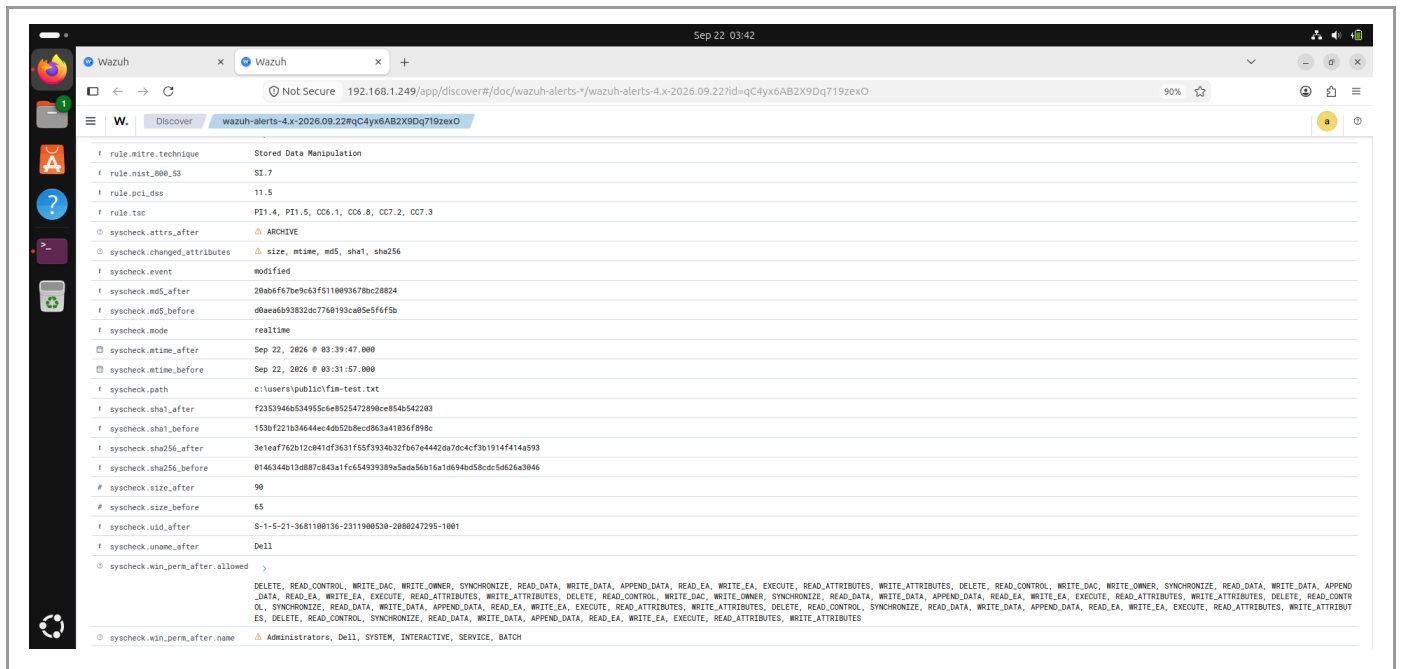


6. Incident 2 — File Modification / File Integrity Monitoring

Detection

A test file named fim-test.txt was created in C:\Users\Public and then modified. Wazuh File Integrity Monitoring detected the change in real time.

The Wazuh event was generated with Rule ID 550, 'Integrity checksum changed', at rule level 7. The event recorded changes including file size, modification time, MD5, SHA1, and SHA256 values.

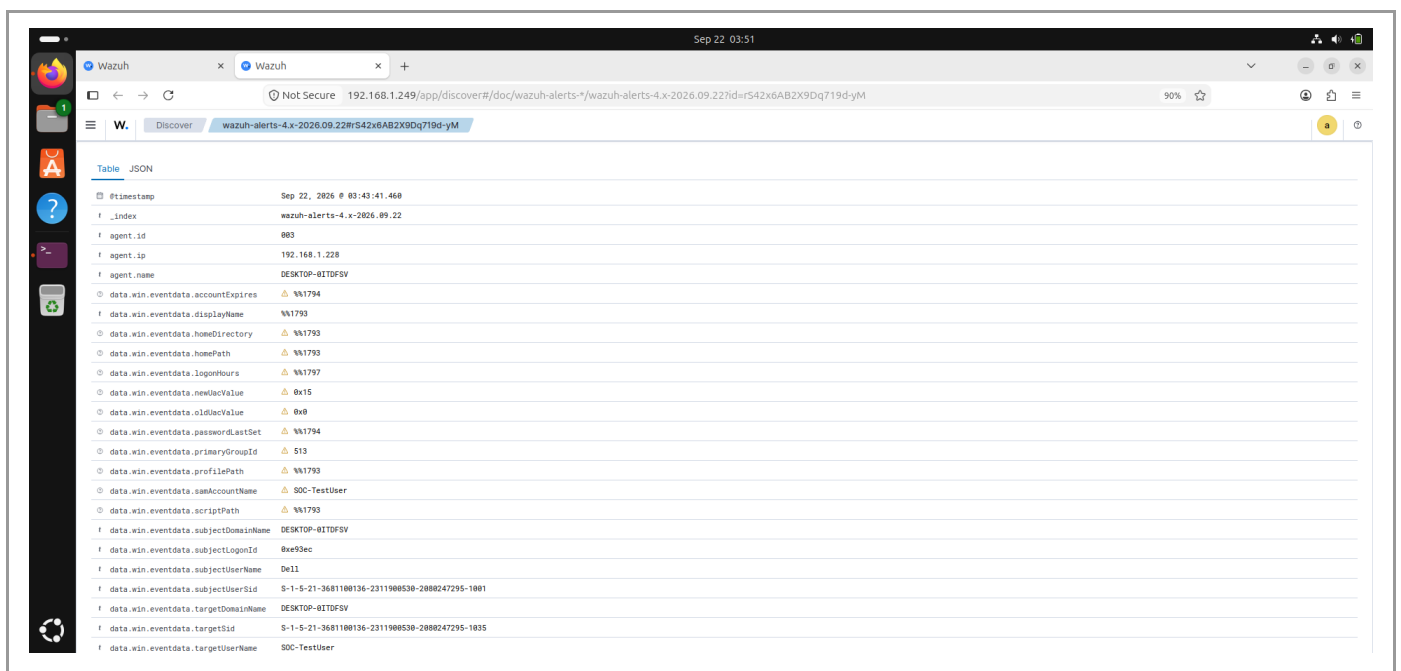
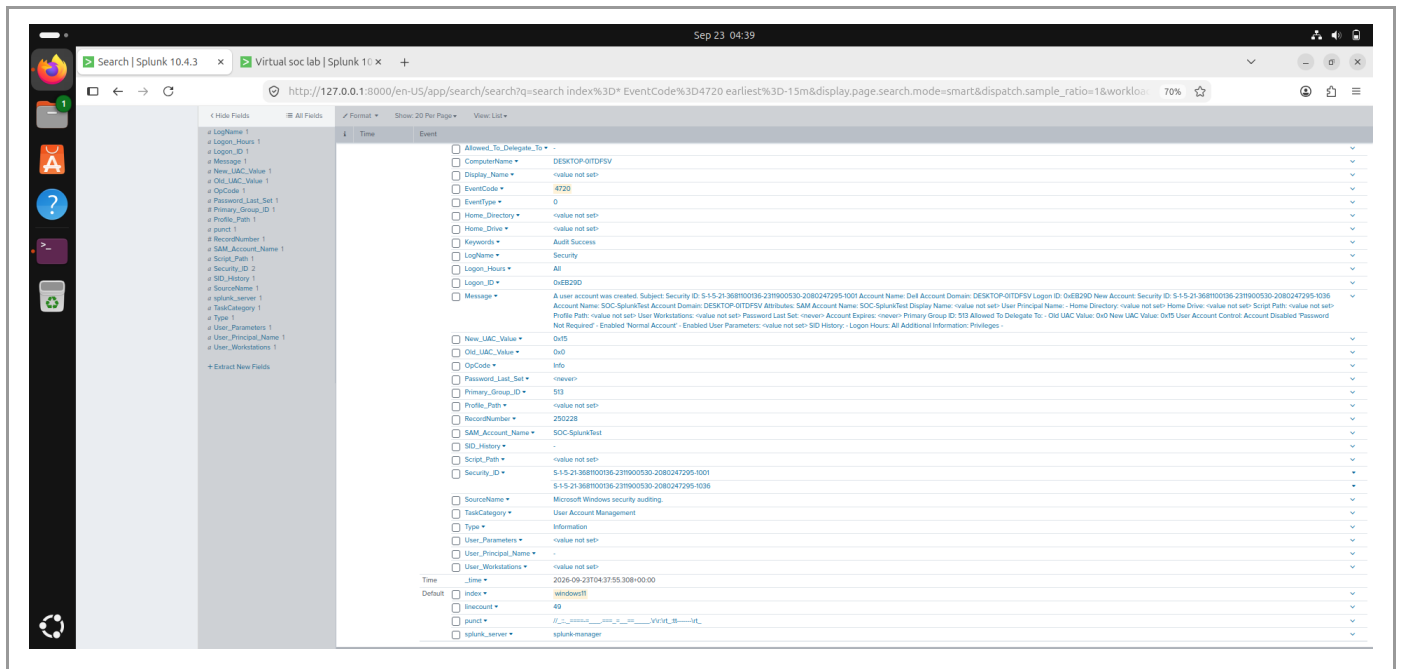


7. Incident 3 — Windows Account Creation

Detection

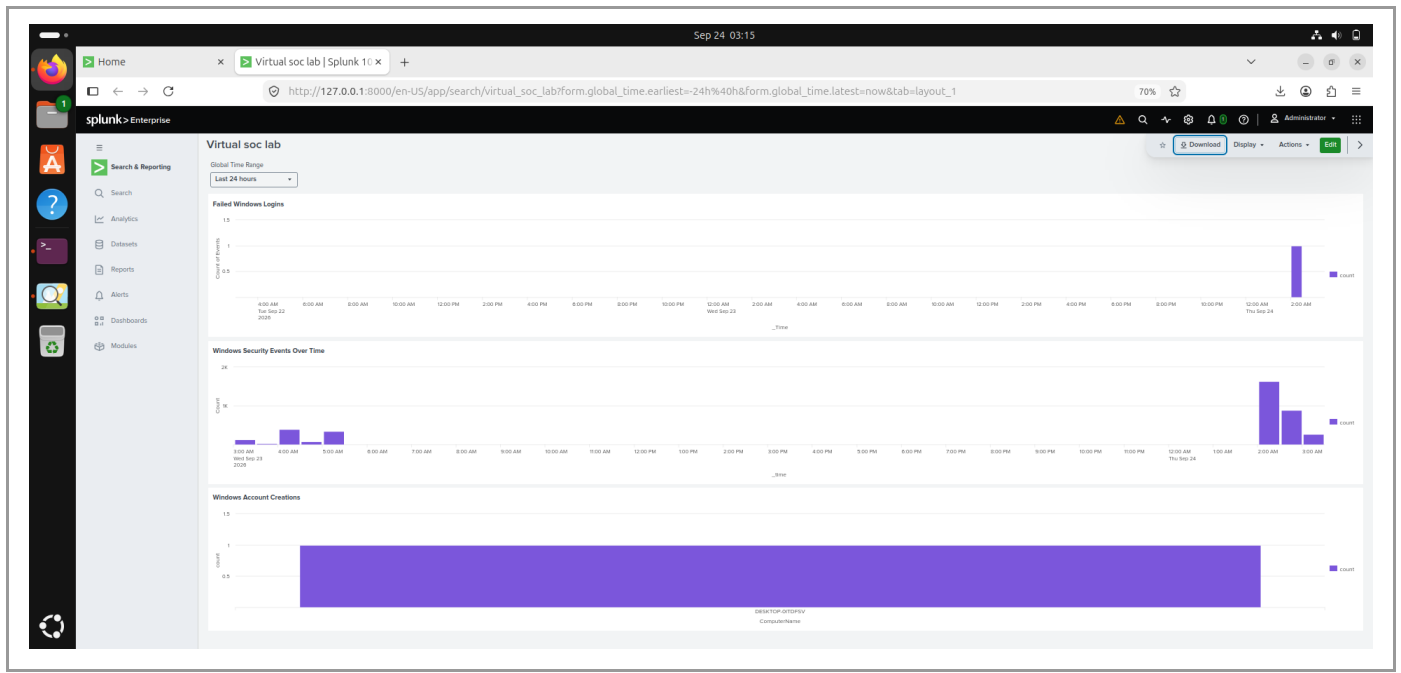
A temporary Windows test account was created to demonstrate account-monitoring capabilities. Windows generated Security Event ID 4720, indicating that a user account was created.

The event was detected in both Wazuh and Splunk. After evidence collection, the temporary test account was deleted.



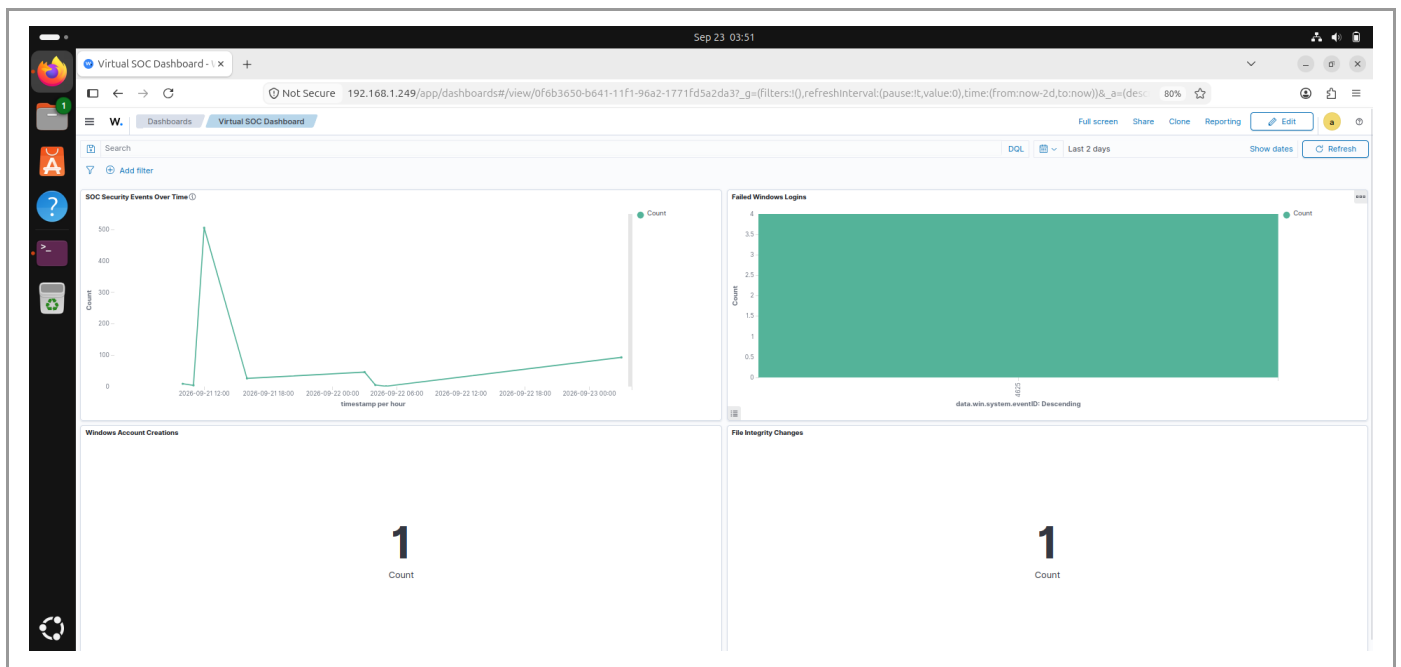
8. Splunk SOC Dashboard

A Splunk dashboard named 'Virtual SOC Lab' was created to provide a simple visual overview of security activity. The dashboard contains panels for failed Windows logins, Windows account creations, and Windows security events over time.



9. Wazuh SOC Dashboard

A Wazuh dashboard named 'Virtual SOC Dashboard' was created with panels for SOC security events over time, failed Windows logins, Windows account creations, and file integrity changes.



10. Incident Correlation

The lab demonstrated how the same controlled security activity can be observed through different monitoring tools. Failed authentication and account-creation events were visible in both Splunk and Wazuh, while file integrity monitoring was handled through Wazuh.

Incident	Windows Event / Rule	Splunk	Wazuh
Failed login	4625 / 60122	Detected	Detected
File modification	FIM / 550	Not detected in this test	Detected
Account creation	4720	Detected	Detected

11. Basic Incident Response

The following basic response workflow was demonstrated conceptually for the lab incidents:

1. Detect the event and identify the affected endpoint.
2. Review the event details and determine what activity occurred.
3. Collect supporting evidence from Splunk and/or Wazuh.
4. Determine whether the activity is expected or requires further investigation.
5. For a suspicious account or authentication event, review the account and authentication context.
6. For file changes, verify whether the modification was authorized.
7. Record the incident, evidence, and response actions.

12. Project Results

The completed lab demonstrated log collection, endpoint monitoring, security event detection, file integrity monitoring, dashboard creation, incident investigation, and basic SOC response workflows in a controlled virtual environment.

Completed practical components:

- Splunk log collection from Windows 11
- Wazuh Windows endpoint enrollment
- Failed-login detection
- File integrity monitoring
- Windows account-creation detection
- Splunk dashboard
- Wazuh dashboard
- Cross-tool incident correlation

13. Conclusion

This project provided practical experience with a basic SOC workflow: collecting endpoint telemetry, detecting security events, investigating alerts, visualizing activity, and documenting incident response. The environment can be extended with additional detection rules, more Windows endpoints, network telemetry, and automated response actions.